



Original software publication

InteractionR: An R package for full reporting of effect modification and interaction

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ABSTRACT

Effect modification and/or Interaction are frequently assessed in epidemiological research. However, in most cases, authors do not present sufficient information for the readers to fully assess the extent and significance of interaction on both additive and multiplicative scale. Also, due to being readily available in most software, the delta method has proliferated in the literature for the estimation of confidence intervals (CIs) for measures of additive interactions; despite its well documented poor performance compared to alternative methods. We introduce *interactionR*, an open-source R package with user-friendly functions that ensures full reporting of effect modification or interaction based on recommended guidelines. In addition to the simple asymptotic delta method, the package also allows for estimation of CIs for additive interaction measures using the variance recovery and percentile bootstrapping methods.

Code metadata

Current code version
Permanent link to code/repository used for this code version
Permanent link to reproducible capsule
Legal code license
Code versioning system used
Software code languages, tools and services used
Compilation requirements, operating environments and dependencies
If available, link to developer documentation/manual
Support email for questions

v0.1.3
<https://github.com/SoftwareImpacts/SIMPAC-2021-115>
<https://codeocean.com/capsule/4102138/tree/v1>
GPLv3
Git
R
R >= 2.10
<https://github.com/epi-zen/interactionR#readme>
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1. Introduction

Although, effect modification analysis is ubiquitous in the epidemiologic literature, its reporting is grossly inadequate [1]. For example, whereas departure from additivity is more relevant to disease prevention and intervention in public health [2], most studies only report interaction on the multiplicative scale since this can be easily extracted from the exponentiated coefficient of the product term of two exposures in a logistic regression model [1].

To improve this situation, Knol and Vanderweele proposed a set of recommendations for the reporting of analyses of effect modification [3] – an improvement on earlier proposals such as STROBE (Strengthening The Reporting of Observational Study) [4]. In their rec-

ommendation, they distinguished effect modification from interaction. Effect modification is when the effect of an exposure on an outcome differs within the strata of a second exposure, while interaction is the causal effect of two exposures together on an outcome [5]. For effect modification analysis with three dichotomous variables: the effect of exposure A on an outcome Y and a putative modifying variable M of this effect, they recommended reporting mainly the following (Table 1a):

- (i) Individual (A on Y (odds ratio (OR)₀₁) and M on Y (OR₁₀)) and joint effect estimates (A and M on Y (OR₁₁)) with confidence intervals (CI), with the stratum of lowest risk (OR₀₀) as the reference

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Table 1
Recommended presentation formats for effect modification and interaction analyses.

^a .			
	A = 0	A = 1	A = 1 vs A = 0 within the strata of M
M = 0	OR ₀₀ (95% CI) [Reference]	OR ₀₁ (95% CI)	OR _{A=1 vs A=0} [M = 0] (95% CI)
M = 1	OR ₁₀ (95% CI)	OR ₁₁ (95% CI)	OR _{A=1 vs A=0} [M = 1] (95% CI)
Effect modification measure on additive scale (95% CI); Effect modification on multiplicative scale (95% CI); ORs adjusted for ...			
^b .			
	A = 0	A = 1	A = 1 vs A = 0 within the strata of M
M = 0	OR ₀₀ (95% CI) [Reference]	OR ₀₁ (95% CI)	OR _{A=1 vs A=0} [M = 0] (95% CI)
M = 1	OR ₁₀ (95% CI)	OR ₁₁ (95% CI)	OR _{A=1 vs A=0} [M = 1] (95% CI)
M = 1 vs M = 0 within the strata of A	OR _{M=1 vs M=0} [A = 0] (95% CI)	OR _{M=1 vs M=0} [A = 1] (95% CI)	
Interaction measure on additive scale (95% CI); Interaction on multiplicative scale (95% CI); ORs adjusted for ...			

^aThough OR (odds ratio) is used here, this could also denote risk ratios (RR) or risk difference (RD).

- (ii) Estimates and CIs of the effect of A on Y within the strata of M and
- (iii) Measures of additive (e.g. Relative Excess Risk due to Interaction, RERI) and multiplicative effect modification with CIs.

If interaction of A and M on Y is of interest, then to report in addition to the above, estimates and CI of the effect of M on Y within the strata of A (Table 1b). Presenting the results this way allows the reader to fully interpret all dimensions of interaction [3].

Fitting a regression model in R [6] will produce some direct estimates required to fill this table, however, estimates such as RERI, OR_{A=1 vs A=0} [M = 1], OR_{M=1 vs M=0} [A = 1] and their CIs are not readily available from the standard output of the regression model and need to be produced with additional methods and coding. These additional steps required may deter the more parsimonious authors from full reporting based on the recommendations presented above.

Complicating things further are the competing methods in the literature for the estimation of CIs around these measures of additive interaction (i.e. RERI, Attributable Proportion (AP) and Synergy Index (SI)). The first such method was the simple asymptotic delta method proposed by Hosmer and Lemeshow [7], which is the most widely used in the literature [8–11] despite its well-documented shortcoming [12, 13]. The delta method estimates the lower and upper bounds of the CIs of measures of additive interaction by imposing an unnatural symmetry on the confidence limits of the underlying effect measure (risk ratio (RR) or OR) that gave rise to the additive interaction measures. This can lead to lower confident limits for measures of additive interaction that are too high, with consequent false positive results [13].

To overcome the shortcomings of the delta method, Assmann et al. [12] suggested a non-parametric bootstrapping approach for CI of RERI, where RERI is estimated each time in a specified number (usually 1000) of bootstrap replications (with replacement) of the original sample. Then the 95% CI of RERI is the 2.5th and 97.5th percentile of the resulting normal distribution. Although, this method has a superior performance if the sample size is large enough [12,13], it falls apart at typical sample sizes at which most observational studies are performed at due to the inevitable sparse cells in some of the bootstrap simulations [14].

Subsequently, the variance recovery ('MOVER') method was proposed by Zou [13]. The method utilizes the asymmetric nature of the confidence limits of RR or OR to recover the variances of these

effect measures for the estimation of CIs for interaction measures. This approach demonstrated better performance to the delta method at typical sample sizes and a comparable performance to bootstrapping at larger sample sizes.

Although, there are R functions [8,9] to estimate relevant measures of additive interactions, they compute CIs using the easily implemented delta method. The alternatives to the delta method described above are buried in the appendices of their publications in spreadsheets, Stata and SAS codes [12,13], making them inaccessible to researchers. The easy availability of the delta method for estimation of CIs of additive interaction measures across the major statistical software may explain its prominence in the literature. In this paper, we introduce 'interactionR' [15], an R package that allows researchers to fully report the results of their interaction analyses (Table 1a or b) in easy steps.

2. Software description

The interactionR is implemented in the R statistical software environment [6]; R is installable from <https://www.r-project.org/>. The main user-facing function of the package, *interactionR()*, accepts the following arguments:

- *model*: a regression model fitted by the user with interaction term for the two exposures under consideration. This may be an object of class glm with a valid link for logistic regression or approximations of risk ratio, class clogit or class coxph. It can also include confounders adjustment as is usually the case.
- *exposure_names*: A character vector of two named binary exposure variables present in the fitted model.
- *ci.type*: A character string ("delta" or "mover") specifying the method to use for the estimation of CI for the measures of additive interaction. Default is "delta".
- *ci.level*: Magnitude of the returned CI level.
- *em*: TRUE, for effect modification assessment. FALSE, for interaction.
- *recode*: If TRUE, recodes the exposures – if at least one of the exposures is protective – such that the stratum with the lowest risk becomes the new reference category when the two exposures are considered jointly (See Knol et al. [16]).

Table 2
Interaction of alcohol and smoking on oral cancers as returned by the ‘*interactionR_table()*’ function.

	smk absent	smk present	Effect of smk within the strata of alc
alc absent	OR [95% CI] 1 [Reference]	OR [95% CI] 2.96 [0.68, 12.91]	OR [95% CI] 2.96 [0.68, 12.91]
alc present	3.33 [0.7, 15.86]	9.04 [2.64, 30.91]	2.71 [1, 7.37]
Effect of alc within the strata of smk	3.33 [0.7, 15.86]	3.05 [1.29, 7.18]	
Multiplicative scale	0.91 [0.15, 5.42]		
RERI	3.74 [−11.43, 21.87]		
AP	0.41 [−0.38, 0.81]		
SI	1.87 [0.65, 5.42]		

Example 1. The joint effect of alcohol and smoking on oral cancer.

Consider the case-control data from Rothman and Keller [17], which studied the joint effect of alcohol and smoking on oral cancers. This dataset was used previously by Hosmer and Lemeshow [18], and Zou [13] and its included in our package as ‘OCdata’ for easy replication of our analysis by the user. It contains the two exposures, alcohol (‘alc’) and smoking (‘smk’), and the outcome, oral cancer (‘oc’) as dichotomous variables (0 or 1, indicating absence or presence respectively). Suppose we are considering interaction and we want to compute the CIs for the measures of additive interaction using the MOVER method, we will start by fitting the following logistic regression model with an interaction term for alcohol and smoking on oral cancer:

$$\text{Logit}(\text{Pr}(\text{OC}_i | \text{Alcohol}, \text{Smoking})) = \alpha + \beta_1 \cdot \text{Alcohol}_i + \beta_2 \cdot \text{Smoking}_i + \beta_3 \cdot \text{Alcohol}_i \cdot \text{Smoking}_i$$

in R as:

```
> m = glm(oc~alc*smk, family = binomial(link = 'logit'), data = OCdata)
```

The fitted ‘m’ object is then passed to the ‘*interactionR()*’ function:

```
> table_object = interactionR(m, exposure_names = c("alc", "smk"), ci.type = "mover", ci.level = 0.95, em=F, recode = F)
```

This returns a list object of class *interactionR* which includes a dataframe containing all the effect estimates that are necessary to fully report effect modification or interaction. This data frame and other components of the list are accessible to the user for further manipulation, if desired. Importantly, the output object is formatted in such a way that the data frame can be processed by the tabling function *interactionR_table()*. An example is:

```
> interactionR_table(table_object)
```

The tabling function will save a publication-ready table as a word document (Table 2) to the user’s working directory, if desired, which corresponds to Table 1b with CIs for RERI, AP and SI estimated with the MOVER method (the set of arguments selected in the original *interactionR()* call). The returned point estimate and CI for RERI (3.74; 95% CI: −11.43, 21.87) (Table 2) is as reported by Zou for this data [13] – If the ‘ci.type’ argument in *interactionR()* had been set to “delta”, the CIs for this trio of interaction measures would be as reported by Hosmer and Lemeshow [18]. In addition to the main function described here, the package also provides standalone functions for each of the CI estimating methods: *interactionR_mover()*, *interactionR_delta()*, and *interactionR_boot()*. The latter implements the percentile bootstrapping of CIs of additive interaction measures as described by Assmann et al. [12] and its described next.

Example 2. Effect of sports participation and smoking on herniated lumbar disc.

To illustrate the *interactionR_boot()* function, consider the case-control data of the effect of sports participation and smoking on herniated lumbar disc examined by Assmann et al. in their analysis. The dataset is available in the package as ‘HDiscddata’ and contains three binary variables: (i) the outcome herniated lumbar disc, ‘h’; and (ii) & (iii) the exposures sports participation, ‘ns’, and smoking ‘smk’. The function accepts the following arguments:

- *model*: A fitted model of class glm. Requires the exposures with interaction term to be listed first before any other covariates/confounders (if applicable).
- *ci.level*; *em*; *recode*: As described for *interactionR()* above.
- *seed*: The random number seed to use for generating the bootstrap samples (for reproducibility). Default is 12345 but can be set to any number.
- *s*: Number of bootstrap resampling. Default is 1000

Again, first, we fit a regression model:

```
> m2 = glm(h ~ns*smk, family = binomial(link = 'logit'), data = HDiscddata)
```

Then, pass the object to the *interactionR_boot()* function as follows:

```
> table_object2 = interactionR_boot(m2, ci.level = 0.95, em = F, recode = F, seed = 12345, s = 1000)
```

This runs a non-parametric bootstrap sample 1000 times with replacements and a percentile CI. The function also returns a list object of class *interactionR* containing all the desired estimates and manipulable by the tabling function - *interactionR_table()* - described earlier. Calling *interactionR_table()* on the returned object produces Table 3, with estimates for the CI of RERI and AP similar to that reported by Assmann et al. for this data.

Furthermore, some base R functions are available to the user to further manipulate some parts of the output object from *interactionR_boot()*. A simple example is:

```
> hist(table_object2$bootstrap)
```

which will produce histograms of the distribution of each of the three bootstrapped parameters (RERI, AP and SI), allowing the user to inspect the overall performance and accuracy of the returned estimates.

3. Impact on academic research

Effect modification and interaction allows us to identify and target high-risk groups in which preventive public health interventions might be most effective [19]. This is crucial for designing preventative strategies and optimally allocating ever-decreasing public health resources [2,5,19]. The *interactionR* package allows researchers to produce publication-ready tables that includes all effect estimates necessary for stakeholders to fully evaluate all dimensions of effect modification and interaction. It also estimates confidence interval for the trio of additive interaction measures using the delta method, variance recovery method, and percentile bootstrapping. The two latter methods are the first implementations in R. The software was used to analyse and report the results of a recently conducted study that evaluated the modification of the effect of tobacco smoking on oral HPV infection by HIV [20]. The study identified people living with HIV who also smoke as a high-risk group to target for any future oral HPV screening development. Furthermore, the package is currently in use by researchers all over the world including at the NIH [personal communication, March 9 & June 28, 2021].

Table 3

Interaction of sports participation (ns) and smoking (smk) on herniation of lumbar disc as returned by the 'interactionR_table()' function.

	smk absent	smk present	Effect of smk within the strata of ns
	OR [95% CI]	OR [95% CI]	OR [95% CI]
ns absent	1 [Reference]	1.88 [1.29, 2.73]	1.88 [1.29, 2.73]
ns present	2.38 [1.27, 4.46]	1.98 [1.12, 3.48]	0.83 [0.3s9, 1.75]
Effect of ns within the strata of smk	2.38 [1.27, 4.46]	1.05 [0.61, 1.83]	
Multiplicative scale	0.44 [0.19, 1.02]		
RERI	-1.28 [-3.63, 0.54]		
AP	-0.65 [-2.21, 0.18]		
SI	0.43 [0.07, 1.41]		

4. Future improvements of the software and limitations

Future iterations of the package will include implementations of the likelihood-based CI estimation for RERI from a linear risk model as proposed by Richardson and Kaufman [14], and updates for any additional robust methods that may be developed. Finally, the examples given in this paper computed additive interaction measures from odds ratios estimated from logistic regression models of case-control datasets used in previous analyses on the subject matter [12,13,18]. This has become the fairly standard approach in the literature [19]. However, the resultant RERI and SI from this approach will be an exaggerated estimate of any underlying additive interaction if the OR from the study does not closely approximate RR [21]. Therefore, users of this approach must ensure that the conditions (such as – but not limited to – rare disease assumption) required for ORs to closely approximate RRs are fully satisfied in their study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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